d The object has only two wheels; this is closer to a motorcycle.
e And you keep going through all the features like the body shape, pedal, and so on, until you arrive at a best guess of the object in the image.

The output of this process is the probability of each class. As you can see in our example, the dog has the lowest probability, 1%, whereas there is an 85% probability that this is a motorcycle. You can see that, although the model was able to predict the right class with the highest probability, it is still a little confused about distinguishing between cars and motorcycles—it predicted that there is a 14% chance this is an image of a car. Since we know that it is a motorcycle, we can say that our ML classification algorithm is 85% accurate. Not bad! To improve this accuracy, we may need to do more of step 1 (acquire more training images), or step 2 (more processing to remove noise), or step 3 (extract better features), or step 4 (change the classifier algorithm and tune some hyperparameters), or even allow more training time. The many different approaches we can take to improve the performance of our model all lie in one or more of the pipeline steps.

That was the big picture of how images flow through the CV pipeline. Next, we'll zoom in one level deeper on each of the pipeline steps.

1.4 Image input

In CV applications, we deal with images or video data. Let's talk about grayscale and color images for now, and in later chapters, we will talk about videos, since videos are just stacked sequential frames of images.

1.4.1 Image as functions

An image can be represented as a function of two variables x and y, which define a two-dimensional area. A digital image is made of a grid of pixels. The *pixel* is the raw building block of an image. Every image consists of a set of pixels in which their values represent the *intensity* of light that appears in a given place in the image. Let's take a look at the motorcycle example again after applying the pixel grid to it (figure 1.15).

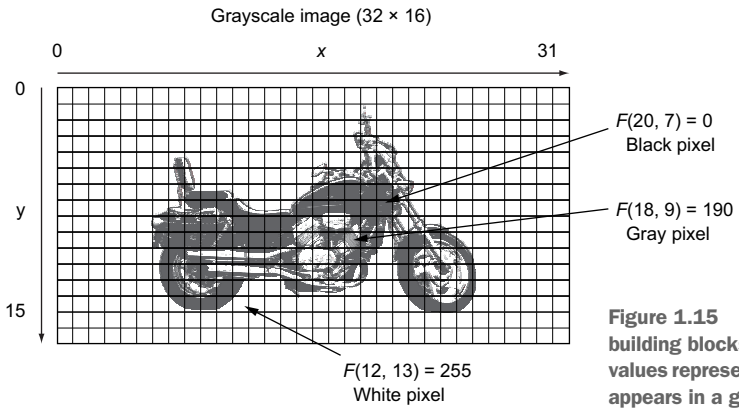

Figure 1.15 Images consists of raw building blocks called *pixels*. The pixel values represent the *intensity* of light that appears in a given place in the image.